

Fundamentals of Wireless Networks & Mobility Management

Architecture, CSMA/CA, Link Characteristics, and Mobile IP Routing

Computer Networking Study Series

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Abstract

This document provides a rigorous, structured overview of wireless local area networks (WLANs), physical link characteristics, Medium Access Control (MAC) protocols, and network-layer mobility management. It incorporates the standard IEEE 802.11 mechanisms (CSMA/CA, RTS/CTS) and evaluates mobile device routing architectures including Mobile IP, Indirect Routing, and Direct Routing.

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1 Elements of a Wireless Network

Wireless networks enable communication between end systems without relying on physical cabling for the final hop. The primary building blocks consist of the following:

Wireless Hosts: End-user systems such as laptops, smartphones, IoT sensors, or desktop systems equipped with a wireless network interface card (NIC).

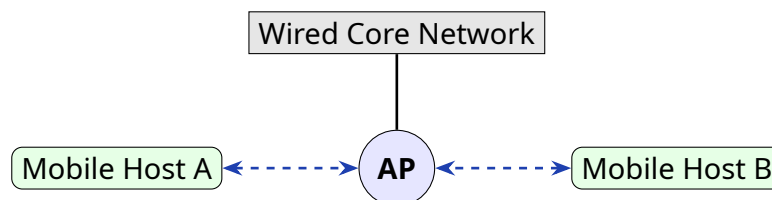
Base Station (Access Point / Cell Tower): A fixed infrastructure node connected to a wired backbone network. It acts as a bridge, relaying packets between the wired backbone and wireless hosts within its coverage area. Examples include Wi-Fi Access Points (APs) and Cellular Base Transceiver Stations (BTS/eNodeB/gNodeB).

Wireless Link: The electromagnetic spectrum (e.g., 2.4 GHz, 5 GHz, or 6 GHz radio frequencies) used to connect hosts to base stations or to other hosts.

Network Infrastructure: The underlying wired backbone, routers, switches, and core network components (e.g., Home/Visited networks) supporting data delivery.

1.1 Architectural Modes

1. **Infrastructure-Based Communication:** Wireless hosts connect directly to a centralized Base Station / Access Point (AP). All traffic destined for other networks or peer hosts passes through the AP, which manages media access and routes packets to the wired core network.
2. **Ad-Hoc Network (Peer-to-Peer):** Nodes communicate directly with one another without a pre-existing infrastructure or central Access Point. Hosts themselves handle routing, address assignment, and network formation dynamically (e.g., MANETs, Bluetooth piconets).



(a) Infrastructure Mode

Figure 1: Infrastructure-based wireless network topology.

2 Wireless Link Characteristics & Physical Layer Issues

Wireless transmission differs fundamentally from wired transmission due to atmospheric interference, multipath propagation, and physical obstacles.

2.1 Path Loss, Attenuation, and High Frequency Fading

Electromagnetic wave propagation obeys the inverse-square law in free space. The path loss model is characterized as:

$$P_r \propto \frac{P_t}{d^\eta} \quad (1)$$

where P_r is the received signal power, P_t is transmitted power, d is distance, and η is the path-loss exponent ($\eta \geq 2$).

- **Attenuation:** As a wireless signal propagates away from the access point, signal strength decays rapidly over distance, resulting in reduced signal-to-noise ratio (SNR) and lower data rates.
- **High Frequency Constraints:** Higher frequency signals (e.g., 5 GHz) offer higher bandwidth but suffer greater attenuation through physical barriers (walls, concrete) compared to lower frequencies (e.g., 2.4 GHz).

2.2 The Hidden Terminal Problem

The **Hidden Terminal Problem** occurs when two wireless transmitters (A and C) are both within range of a shared receiver (B), but cannot detect each other due to physical obstacles or geographic distance.

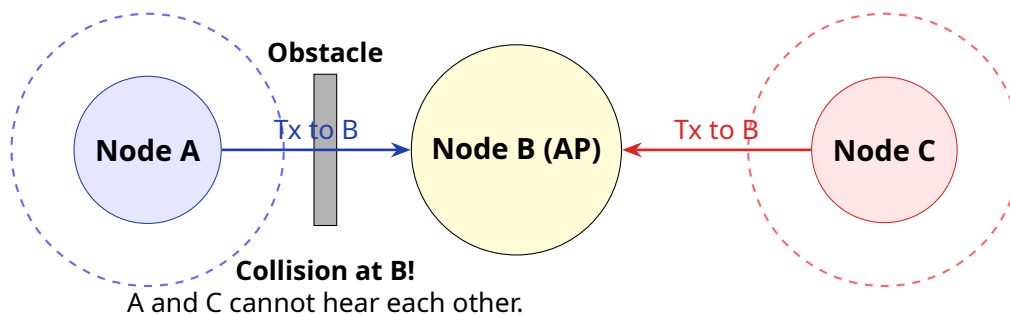


Figure 2: The Hidden Terminal Problem: Nodes A and C simultaneously transmit to B because neither detects the other's signal.

3 IEEE 802.11 (Wi-Fi) & CSMA/CA Protocol

IEEE 802.11 networks organize architecture around a **Basic Service Set (BSS)**, which consists of an Access Point (AP) and associated wireless nodes operating on assigned frequency channels.

3.1 Why CSMA/CD Cannot Be Used in Wireless

Standard Ethernet uses Carrier Sense Multiple Access with Collision Detection (CSMA/CD). However, CSMA/CD fails in wireless environments because:

1. **High Dynamic Range:** A transmitter's own signal overwhelms its local receiver hardware, making it extremely difficult to detect incoming low-power collisions while transmitting.
2. **Hidden Terminals:** A sender might sense the channel as clear locally, while a collision is actively occurring at the receiver end.

3.2 CSMA/CA Mechanics

To mitigate collisions, IEEE 802.11 implements **Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)** along with explicit Link-Layer Acknowledgments (ACK).

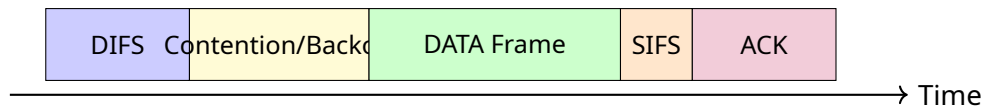


Figure 3: Basic CSMA/CA Frame Transmission Sequence.

3.2.1 Step-by-Step CSMA/CA Algorithm

1. **Sense Channel:** If the channel is sensed idle for a duration equal to **DIFS** (Distributed Inter-Frame Space), the node proceeds.
2. **Backoff Timer:** If the channel is sensed busy, the node selects a random backoff value from a contention window $[0, CW]$. The timer decrements while the channel is idle and freezes when the channel is busy.
3. **Transmit Data:** When the timer reaches zero, the sender transmits the complete frame.
4. **Receiver ACK:** If received correctly, the destination waits for a duration equal to **SIFS** (Short Inter-Frame Space) and responds with an ACK frame.

Key Property: SIFS vs. DIFS

SIFS < DIFS: Because SIFS is shorter than DIFS, control frames (like ACKs, RTS, and CTS) always take priority over pending data frame transmissions, preventing interruption from other competing nodes.

3.3 Virtual Carrier Sensing via RTS/CTS

To prevent collisions caused by hidden terminals, IEEE 802.11 provides an explicit **Request-to-Send (RTS)** and **Clear-to-Send (CTS)** reservation mechanism.

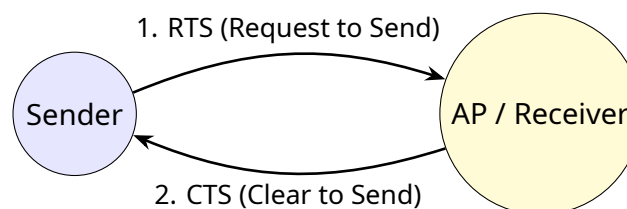


Figure 4: RTS/CTS reservation handshake.

- **RTS Frame:** The sender broadcasts an RTS containing the required transmission time duration.
- **CTS Frame:** The receiver replies with a CTS frame. All nodes hearing the CTS defer their transmissions for the specified duration (setting their **Network Allocation Vector / NAV**).
- **Verification:** Successful CTS reception confirms the channel is clear at the receiver's end, allowing safe data frame transmission.

4 Mobility Management in Wireless Networks

4.1 The Spectrum of Mobility

Network mobility is categorized based on the degree of physical movement and connection continuity across provider boundaries:

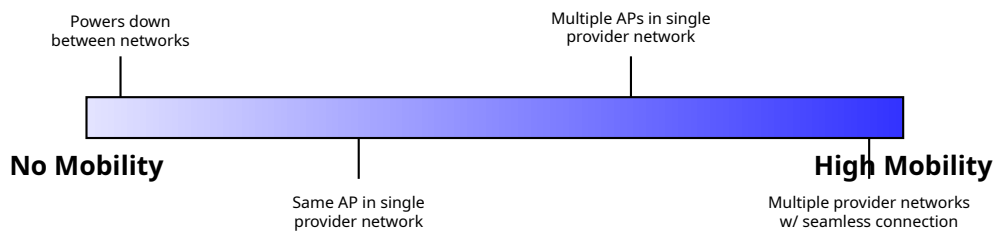


Figure 5: The Spectrum of Mobility (From static access to seamless roaming).

4.2 Mobile IP Terminology & Architecture

When a mobile node (MN) moves between networks, its IP address poses a dilemma: **IP addresses encode location for routing purposes**. If an MN changes its IP address, active TCP connections drop. Mobile IP solves this by decoupling identity from physical location.

Permanent Identity (Home Address): A static IP address assigned to the mobile device in its home network, remaining unchanged regardless of location.

Home Agent (HA): An entity inside the home network that tracks the mobile device's current location and intercepts packets addressed to the device's Home Address.

Foreign / Visited Network: The foreign subnet that the mobile node attaches to when away from home.

Care-of-Address (CoA): A temporary, dynamically assigned IP address in the visited network used to route packets to the mobile node.

Tunneling: The process where the Home Agent encapsulates original IP packets inside an outer IP packet addressed directly to the Care-of-Address.

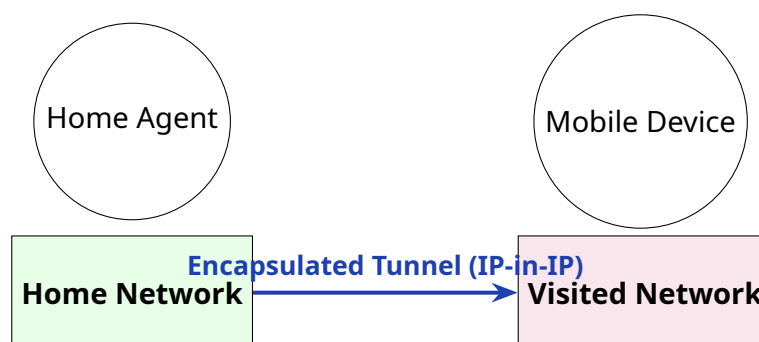


Figure 6: Mobile IP tunneling architecture forwarding packets to a visited network.

5 Routing to Mobile Nodes: Indirect vs. Direct Routing

5.1 Indirect Routing

In **Indirect Routing**, the correspondent node (sender) remains unaware of the mobile node's mobility and sends traffic directly to the mobile node's permanent IP address.

1. Sender transmits packets to the Mobile Node's permanent Home Address.
2. The **Home Agent** intercepts these packets at the home network.
3. The Home Agent encapsulates the packet and routes it through a **tunnel** to the foreign Care-of-Address (CoA).
4. The packet is decapsulated in the visited network and delivered to the mobile node.

5.1.1 Drawback: Triangle Routing

Indirect routing suffers from **Triangle Routing Inefficiency**. If the sender and mobile receiver are physically adjacent, packets must still travel across the globe to the Home Agent before being tunneled back to the visited network, introducing severe propagation delay and network overhead.

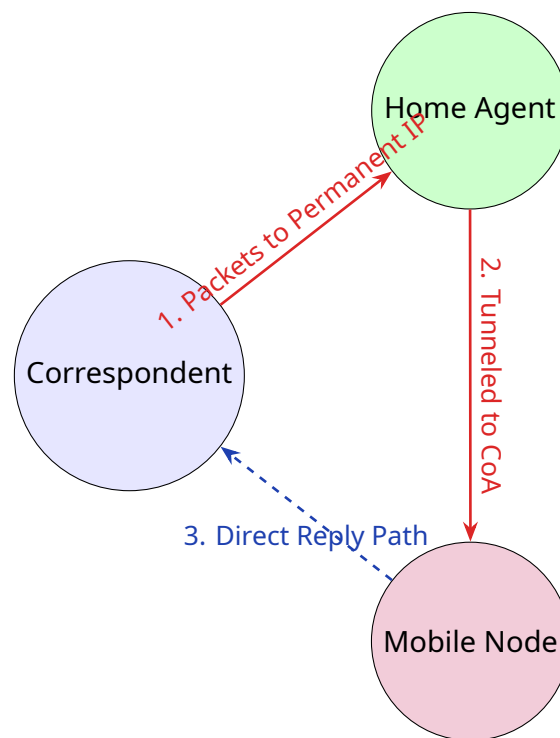


Figure 7: Triangle Routing: Direct replies bypass the Home Agent, but incoming path suffers routing delays.

5.2 Direct Routing

Direct Routing eliminates triangle routing by letting the sender query the mobile node's current location prior to data exchange.

1. The sender queries the Home Agent to obtain the mobile node's current **Care-of-Address (CoA)**.
2. The Home Agent returns the CoA to the sender.
3. The sender directly establishes a connection and transmits data straight to the Care-of-Address in the visited network.

Trade-off: While direct routing avoids latency issues, it introduces protocol setup overhead, initial lookup latency, and requires the sender to support mobile location lookup mechanisms.

5.3 Comparison Matrix

Table 1: Comparison of Indirect and Direct Mobility Routing Approaches

Feature	Indirect Routing	Direct Routing
Sender Transparency	Fully transparent to sender	Requires sender modification/lookup
Routing Efficiency	Poor (Triangle Routing problem)	Optimal (Direct short path)
Initial Setup Overhead	Very Low	Higher (CoA query phase)
Tunneling Point	Between Home Agent & Visited Net	Between Sender & Visited Net
Handover Complexity	Handled centrally by Home Agent	Handled dynamically by Sender & Nodes

6 Verification & Self-Assessment Checklist

To verify comprehension of these wireless concepts, complete the following checks:

- ☐ **Verification 1 (Hidden Terminal):** Explain why RTS/CTS prevents collisions at the receiver even when two senders cannot hear each other.
- ☐ **Verification 2 (CSMA/CA Timing):** Confirm why SIFS < DIFS guarantees priority for control frames over data transmission initiation.
- ☐ **Verification 3 (Mobile IP):** Identify why a mobile device retains its permanent IP address instead of relying solely on dynamic DHCP reconfiguration when roaming across networks.